SIMATS ENGINEERING

ASSESSMENT TOOL III— Interactive Dashboard Project

Course: Data Handling and Visualization	**Code: DSA0601**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No : 192424322** **Name: M.Arshini**

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods, applied through the design of interactive dashboards. (BL4)

Activity Tool : Interactive Dashboard Project

Weightage: 20%

Code of Conduct:

I **M.Arshini 192424322** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Arshini

Assessment Criteria

Criteria	Dashboard Design and Understanding 25%	Visual Analysis and Interpretation 25%	Application of Dashboard Design Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Interactive Dashboard Project — Assessment Tool

Tableau Practical Questions with Datasets

Instructions: Answer all five questions. Use Tableau to construct the required interactive dashboards, analyze the patterns, compare design choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Building a Sales Performance Dashboard

A retail company records regional sales performance across product categories. Using Tableau, connect to the dataset below and build an interactive dashboard containing: a bar chart of Sales by Region, a line chart of Sales trend by Month, and a map view of Sales by Region. Add a Region filter that acts as a global filter across all views. Analyze which region and category combinations drive the highest sales, and explain how the dashboard layout supports quick comparison across regions.

Data:

S.No	Region	Category	Sales
1	North	Furniture	32000
2	South	Furniture	28500
3	East	Furniture	41000
4	West	Furniture	25500
5	North	Technology	51000
6	South	Technology	47500
7	East	Technology	62000
8	West	Technology	39500
9	North	Office Supplies	21000
10	South	Office Supplies	19500
11	East	Office Supplies	27000
12	West	Office Supplies	17500

Expected Tableau tasks: data connection and preparation, bar chart and line chart worksheets, map visualization, global filter action, dashboard layout and formatting, interpretation of regional and *category patterns.*

Objective

To analyze sales performance across different regions and product categories using interactive Tableau visualizations.

Dashboard Components

Bar chart showing Sales by Region

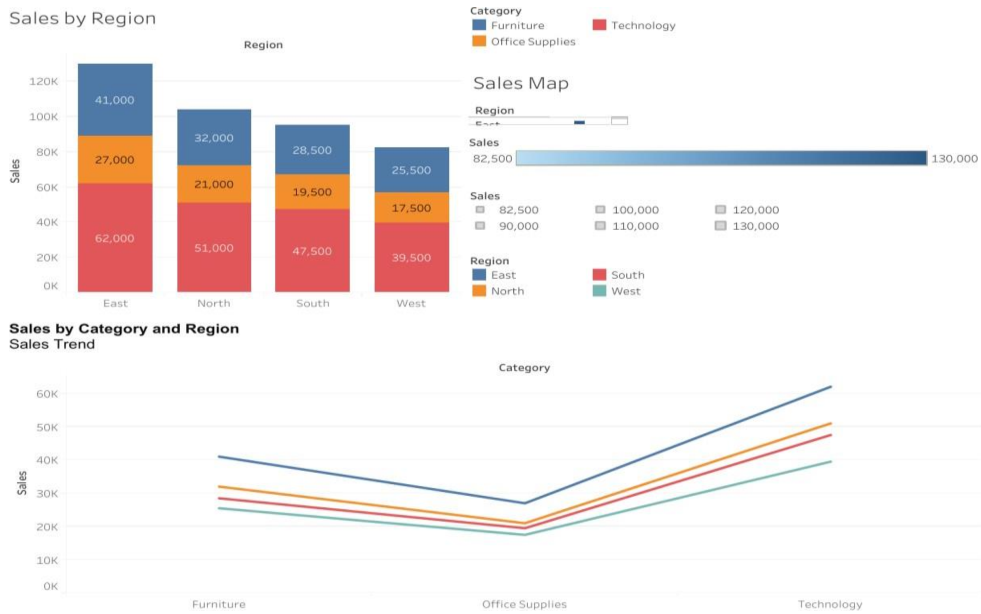
Line chart showing sales comparison using the available data

Regional sales view/map

Region filter applied across the dashboard

Analysis

The East region has the highest total sales of 130,000, followed by North with 104,000, South with 95,500, and West with 82,500. Technology is the best-performing category overall. The highest region-category combination is East–Technology with sales of 62,000. The dashboard allows quick comparison of regional performance using the bar chart and interactive Region filter.



Conclusion

The dashboard provides an easy way to compare sales performance across regions and categories. The interactive filter helps users focus on individual regions.

Q2. KPI Dashboard with Parameters

An HR analytics team wants to monitor employee performance across departments. Using the dataset below, build a Tableau dashboard containing KPI summary cards (Average Rating, Total Employees, Average Attendance) and a bar chart of Average Rating by Department. Add a parameter control that lets the user switch the bar chart measure between Average Rating and Average Attendance. Explain how parameters improve the dashboard's flexibility compared to using separate static charts.

Data:

S.No	Department	Rating	Attendance
1	Sales	7.8	92
2	Sales	8.1	95
3	Marketing	7.5	88

S.No	Department	Rating	Attendance
4	Marketing	7.9	90
5	Engineering	8.6	96
6	Engineering	8.9	97
7	Support	7.2	85
8	Support	7.4	87
9	HR	8	93
10	HR	8.3	94

Expected Tableau tasks: calculated fields for KPI cards, parameter creation, calculated field driven by parameter, bar chart worksheet, KPI card formatting, dashboard assembly, interpretation of flexibility gained from parameters.

Objective
To monitor employee performance using KPI cards and an interactive parameter-controlled bar chart.

Dashboard Components
Average Rating KPI

Total Employees KPI

Average Attendance KPI

Department-wise bar chart

Parameter to switch between Average Rating and Average Attendance

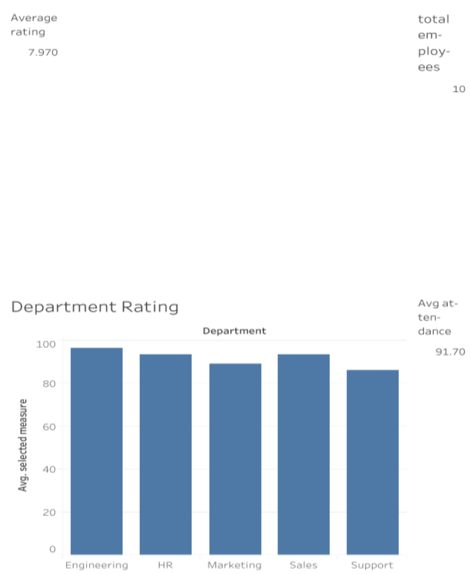
KPI Results
Average Rating = 8.02

Total Employees = 10

Average Attendance = 91.7%

Analysis
Engineering has the highest average rating of 8.75, while Support has the lowest average rating of 7.30. Engineering also has the highest average attendance of 96.5%, while Support has the lowest at 86%.

Parameter Explanation
The parameter improves dashboard flexibility by allowing the user to switch between Average Rating and Average Attendance in the same bar chart. This avoids creating separate static charts and saves dashboard space.



Conclusion
The KPI dashboard provides a quick summary of employee performance and allows users to interactively compare different performance measures.

Q3. Drill-Down Dashboard with Dashboard Actions

A college wants to analyze student enrollment across Program and Specialization levels. Using the hierarchical dataset below, build a Tableau dashboard with a summary chart at the Program level and a detail chart at the Specialization level. Configure a filter action so that clicking a bar in the Program chart updates the Specialization chart to show only the relevant rows. Analyze how the drill-down interaction improves exploration compared to showing all specializations at once, and note any limitations of this approach.

Data:

S.No	Program	Specialization	Enrollment
1	CSE	AI/ML	65

2	CSE	Cyber Security	40
3	CSE	Core CSE	90
4	ECE	VLSI	35
5	ECE	Communications	50
6	MECH	Robotics	30
7	MECH	Thermal	45
8	MECH	Design	28

Expected Tableau tasks: hierarchy or two related worksheets, filter action configuration between sheets, dashboard interactivity testing, interpretation of drill-down behavior and its limitations.

Objective

To analyze student enrollment at Program and Specialization levels using an interactive drill-down dashboard.

Dashboard Components

- Program-level enrollment bar chart
- Specialization-level enrollment bar chart
- Dashboard Filter Action

Results

Program	**Enrollment**
CSE	195
MECH	103
ECE	85

CSE has the highest enrollment with **195 students**.

Analysis

CSE has the highest total enrollment, followed by MECH and ECE. Within CSE, Core CSE has the highest enrollment with 90 students. The dashboard filter action allows users to click a program and view only its related specializations.

Conclusion
The drill-down interaction reduces visual clutter and makes detailed exploration easier. However, users need to select a program to view its specializations, making simultaneous comparison of all specializations less convenient.

Q4. Multivariate Analysis Dashboard

A marketing study records advertising expenditure, website visits, and sales for 15 observations. Build a Tableau dashboard with a scatterplot of Advertising versus Sales, using color or size encoding to represent Website Visits, along with a trend line. Add a second view (box plot or histogram) summarizing the distribution of Sales. Analyze the strength and direction of the relationship shown in the scatterplot and discuss whether the additional Website Visits encoding adds insight or clutters the dashboard.

Data:

S.No	Advertising	Visits	Sales
1	10	120	48
2	12	135	50
3	14	150	53
4	16	165	55
5	18	180	58
6	20	195	61

7	22	210	63
8	24	225	66
9	26	245	69
10	28	260	71
11	30	275	74
12	32	290	77
13	34	305	80
14	36	320	83
15	38	340	86

Expected Tableau tasks: scatterplot with trend line, color/size encoding for a third variable, secondary distribution view, dashboard layout with legends, relationship analysis and critique of visual clutter.

Objective
To analyze the relationship between Advertising, Website Visits, and Sales using a scatter plot and sales distribution.

Dashboard Components
Advertising vs Sales scatter plot

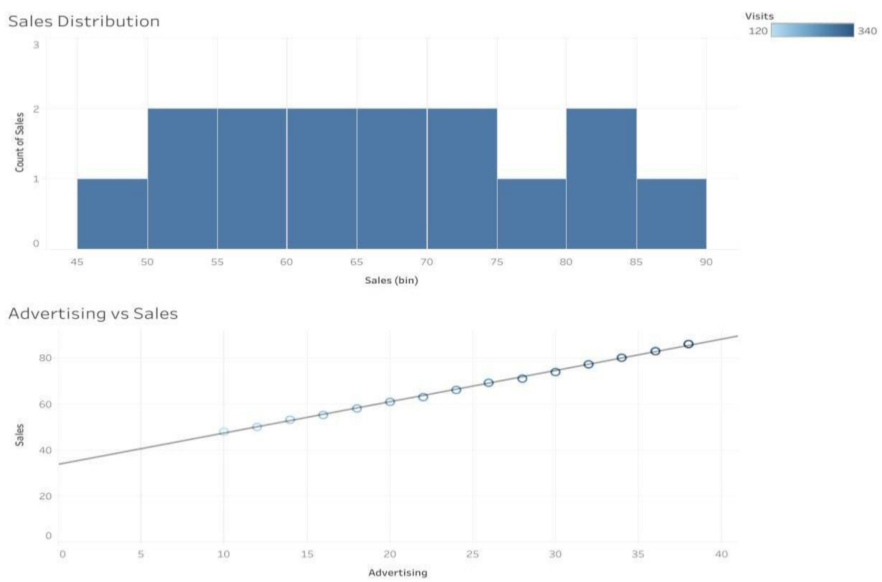
Website Visits represented using color

Linear trend line

Sales distribution histogram

Analysis
The scatter plot shows a strong positive relationship between Advertising and Sales. As advertising expenditure increases, sales also increase. The upward trend line confirms this positive relationship. Website Visits provide additional information because visits also increase as advertising expenditure increases.

Visual Evaluation
Using Website Visits as color adds another dimension to the analysis and helps identify the pattern in the observations. However, using too many visual encodings could make the scatter plot difficult to understand. Therefore, using color alone is sufficient.



Conclusion

The dashboard effectively communicates the positive relationship between advertising expenditure and sales while providing additional insight through website visits.

Q5. Misleading Dashboard and Redesign

A college reports the following pass percentages for seven departments. Build two Tableau dashboards: (1) a misleading version using a truncated y-axis, an inappropriate color scheme, and exaggerated visual emphasis on one department, and (2) a redesigned version using an appropriate baseline, ordered categories, consistent color encoding, and clear labels. Compare the two dashboards and explain how the first design can mislead viewers, and why the redesigned dashboard communicates the data more honestly.

Data:

S.No	Department	Pass_Percentage
1	CSE	91
2	ECE	93
3	EEE	92
4	MECH	89
5	CIVIL	90
6	AIDS	95
7	IT	94

Objective

To demonstrate how inappropriate visualization choices can mislead viewers and how proper dashboard design improves accurate communication.

Misleading Dashboard

The misleading dashboard uses:

- Truncated Y-axis starting at **85**
- Inappropriate/multiple colors
- Excessive visual emphasis on one department

Redesigned Dashboard

The redesigned dashboard uses:

- Y-axis starting from **0**
- Departments ordered from highest to lowest
- Consistent color encoding
- Clear percentage labels

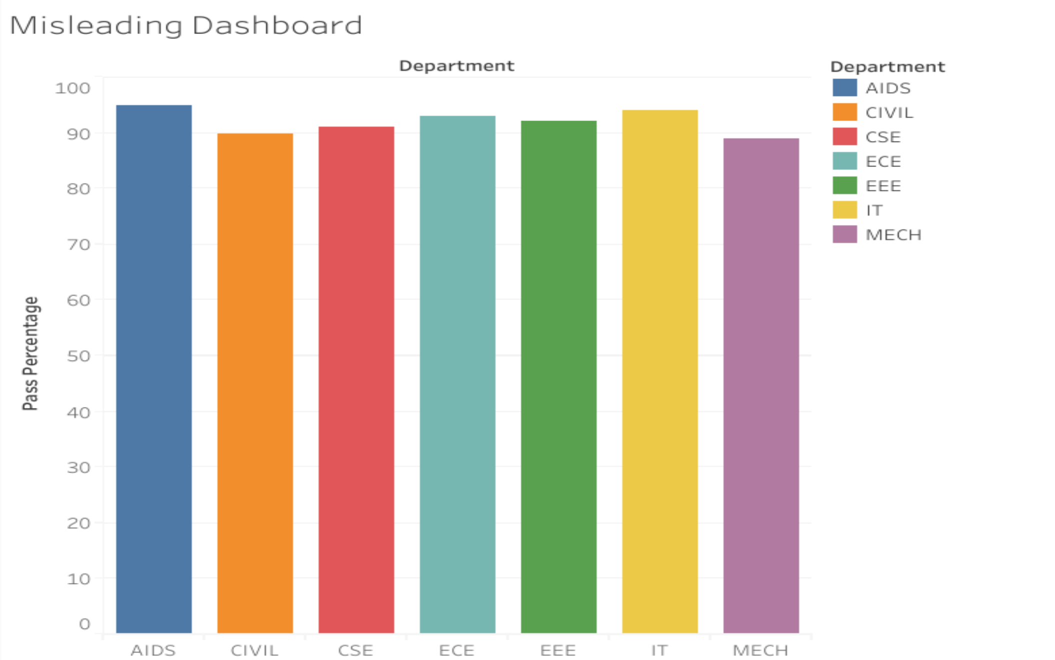
Data Results

Department	Pass Percentage
AIDS	95%
IT	94%
ECE	93%
EEE	92%
CSE	91%
CIVIL	90%
MECH	89%

Analysis

The misleading dashboard exaggerates the differences between departments because the Y-axis is truncated. Multiple unrelated colors and excessive emphasis on one department can also influence the viewer's perception.

The redesigned dashboard presents the data more honestly by using a suitable baseline, ordered categories, consistent colors, and clear labels. The actual difference between the highest and lowest departments is only **6 percentage points**.



Conclusion

The redesigned dashboard provides a clearer and more truthful representation of the data. Appropriate visualization choices help viewers understand the actual differences without unnecessary visual distortion.